

Hayden Choi

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Education

McMaster University

Honours Computer Science Co-op, Bachelor of Applied Science (GPA: 3.9/4.0)

Expected Dec 2027

Hamilton, ON

Experience

Software Engineer Intern

May 2026 – Present

Trend Micro

Ottawa, ON

- Built a behavioral analysis framework that ranks **370+** security detection rules by quality, scoring each on whether fleets remediate or dismiss it across **8M+** decisions from **2,000+** tenants on a **Spark/Trino Iceberg** lakehouse
- Engineered a dismissal-based early-warning signal that flags exposures fleets silently ignore, computed from fleet-wide remediation patterns and back-tested against a critical risk it caught **17 months** before manual escalation
- Eliminated volume-inflation bias by sourcing rankings from a human-action decision ledger instead of **7B+** row raw event tables, weighting by distinct-customer breadth where one tenant drove **72%** of a rule's volume
- Architected and shipped a standalone **EKS** reader service with no static credentials, owning its **Terraform** footprint (**EKS Pod Identity**, least-privilege **IAM**), run on-demand and on a schedule to rank underperforming rules
- Audited a pre-release **gRPC** connector SDK, discovering **7+** packaging, versioning, and API-contract defects

Software Engineer Intern

Dec 2025 – Apr 2026

Knockri

Toronto, ON

- Architected an end-to-end **ReAct AI** agent on IBM Watsonx Orchestrate that autonomously resolves customer support tickets, reducing ticket volume by **83%** and external support costs by **60%**
- Reduced transcription tool latency **74%** by resolving **N+1 GraphQL** queries and refactoring data-fetching architecture
- Engineered **Knockri CLEAR**, a secure proctoring suite using Django and React that monitors and validates **10k+** assessment sessions across Fortune 500 and government clients
- Implemented a **Heartbeat API** via **GraphQL/UUID4** to restrict **7,000+** duplicate launches via backend locking
- Safeguarded production stability by writing **Cypress E2E** and **Django mock tests** for automated **CI/CD** regression

Software Engineer Intern

May 2025 – Aug 2025

Brighter Signals BV

Markham, ON

- Built a real-time signal processing pipeline for an **F1 plank wear detection** prototype, applying baseline calibration at ingestion to denoise raw sensor data and improve detection accuracy by **64%**
- Developed **5+** specialized GUIs for investor demos that secured **\$1.6M+** in funding for **MVP** development
- Shifted **Signal Pre-processing** to embedded firmware to automate filtering, reducing frontend compute load by **76%**
- Co-authored an **IROS 2025** paper and named a **Best Poster Finalist** for research on data-rich tactile sensors

Software Engineer Intern

May 2024 – Sept 2024

Mana Immigration

Markham, ON

- Architected **6+** full-stack Flutter apps for international clients using a Factory Pattern for reusable UI components
- Boosted app responsiveness by **50%** by implementing **Deferred Loading** for complex navigation modules
- Engineered a **Discovery Engine** via **Spotify REST APIs** that bypassed algorithmic bias for genre-isolated playlists

Projects

ColdApproach AI | *Next.js, FastAPI, AWS (SES, SQS, Lambda), OpenAI, Hunter.io*

GitHub | Website

- Built an AI outreach platform with **FastAPI** and **Next.js** using **Hunter.io** discovery and a two-pass **GPT** pipeline
- Engineered an async batch pipeline with **AWS SQS** to process up to **20** URLs through scraping and generation stages
- Implemented **EventBridge Scheduler** with **AWS Lambda** for delayed email delivery via a provider factory pattern
- Designed a **Next.js** and **TypeScript** frontend with **Zustand** persistence and **React Query** polling for batch tracking

SyllaScan | *React, Python, OpenAI, AWS, FastAPI, Google Calendar API*

GitHub

- Developed an **NLP pipeline** using **FastAPI** to parse unstructured PDF syllabi into validated **JSON** schemas
- Applied **Prompt Engineering** strategies to extract complex date-event pairs with high accuracy via **GPT-4**
- Integrated **Google OAuth 2.0** and **Calendar API** to automate bulk event synchronization for academic schedules
- Built a responsive **React** interface for real-time file uploads and interactive schedule previews before final sync

Technical Skills

Languages: Python, JavaScript, Java, TypeScript, Ruby, C++, C, SQL

Libraries/Frameworks: Django, React, PyTorch, Scikit-learn, FastAPI, PySpark

Databases/Cloud: AWS (S3, Glue, Athena, EMR, SES, SQS), Azure, Apache Iceberg, Trino, MongoDB

Tools/Methodologies: Git, Docker, Kubernetes, Terraform, Linux, GraphQL, CI/CD, Agile